

NON SHIVERING THERMOGENESIS AND IMPLICATION OF THE
THYROID IN COLD LABILE AND COLD RESISTANT POPULATIONS
OF THE GOLDEN SPINY MOUSE (*ACOMYS RUSSATUS*).

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The golden spiny mouse is dependent on non shivering thermogenesis (N.S.T.) for thermoregulation at cool ambient temperatures. Mice from the shores of the Dead Sea (Ein-Gedi, EG-mice) lose body heat when exposed to 6°C. Their rate of cooling is linearly correlated to the magnitude of N.S.T. This is true for mice acclimated to 28°C, born in the laboratory or freshly captured. Mice from the high mountains of South Sinai (S-mice) resist cooling under the same conditions and their N.S.T. is about twice that of EG-mice. EG-mice did not acclimate to cold. However thyroxine injections made them cold resistant and their N.S.T. rose to values close to that of S-mice. Light and electronmicroscopy of the thyroids in mice acclimated to 28°C and exposed to 6°C, or injected with TRH suggested intense activity in S-mice and little activity in EG-mice.

Small mammals exposed to cold are not dependent for their survival on non-shivering thermogenesis (N.S.T.); shivering replaces it. This is the case in animals not acclimated to cold, or upon blockage of the sympathetic-adrenomedullary system responsible for the activation of N.S.T. [8, 7]. A much closer dependency was found in a murid species native to the extreme deserts of the Near East and Arabia, namely, the golden spiny mouse, *Acomys russatus* (weight around 60g). Mice brought from Ein-Gedi, on the shores of the Dead Sea (EG-mice), were acclimated to 28°C from one to several months. (For techniques of trapping and breeding see [10]). Individual mice were exposed, in a metal cage with bedding and food, to ambient temperature (T_a) of 6°C in a cold room. Their body (rectal) temperature was measured with a thermistor

(YSI) every hour or so (T_b , or T_r). In every case a decline of T_b was seen (compare to Fig. 2). Each mouse was marked, and after a recovery period of 2 weeks at 28°C, its maximal N.S.T. was measured at 31°C by injecting norepinephrine (1.5mg/Kg, s.c.) and recording the increase in oxygen consumption. A Beckman G₂-O₂ analyzer was used. The experimental procedure was similar to that of Heldmaier [5]. The Na-Pentobarbital dose being only 35 to 60 mg/Kg. The acclimated EG-mice, or those born and bred at 28°C in our laboratory, showed a straight line relationship between the magnitude of N.S.T. measured and the rate of decrease in T_b when exposed to 6°C (Fig. 1) [3]. The magnitude of N.S.T. is presented here as the relative increase in $\dot{V}_{O_2}$ over the resting metabolic rate (R.M.R.), measured in the same experiment before norepinephrine (NE) injection. The rate of decrease in T_b - as the time (in hours) for T_b to decrease by 5°C. The relationship described in Fig. 1 suggests little participation of other thermogenic mechanisms in the resting spiny mouse. This may be contrasted with the more complex situation described in rats [6]. In order to see whether mice in the field have a similar dependency, four mice trapped at the end of winter in Ein Gedi were examined: Two were injected with NE for N.S.T. measurement 24 hours after capture and exposed to 6°C 24 hours later. The second pair was exposed to 6°C first and had their N.S.T. measured afterwards. The resulting straight line (Fig. 1) points to the same conclusion. The slope is steeper in this case because the freshly captured mice have probably